Early-Pandemic COVID-19 Epidemiologic Analysis

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```
library(ggplot2)
```

1 Study Data

This study uses a 10,000-record sample of reported COVID-19 cases restricted to the early pre-vaccine study period from March 15 through December 10, 2020.

The dataset for this study was acquired from the CDC COVID-19 Case Surveillance Public Use Data through the CDC Socrata API.

The study dataset is a computational subsample of and was constructed by randomly selecting 20 calendar dates within the study period using a fixed random seed (42). For each selected date, up to 500 records were retrieved from a contiguous block of surveillance records, with the starting position selected using a date-specific random seed. Records were ordered by their internal Socrata identifier before retrieval.

The resulting dataset contains 10,000 surveillance records and 12 variables. Since sampling was performed by date and contiguous record blocks rather than by simple random sampling across all eligible cases, the sample should not be considered representative of all reported COVID-19 cases during the study period.

```
# Load COVID-19 surveillance sample
covid <- read.csv(
    "Data/covid_early_pandemic_sample_10k.csv",
    stringsAsFactors = FALSE
)

cat(sprintf("Rows: %s | Columns: %s\n",
    format(nrow(covid), big.mark = ","),
    format(ncol(covid), big.mark = ",")))
```

```
Rows: 10,000 | Columns: 12
```

2 Analytic Cohort

The analysis is restricted to cases with known exposure and outcome status. Exposure is defined by `medcond_yn`, indicating whether a pre-existing medical condition was reported. The outcome is defined by `hosp_yn`, indicating a hospitalization was reported.

`medcond_yn` and `hosp_yn` are coded as {`Yes`, `No`, `Unknown`, `Missing`}. Only `Yes` and `No` records for both variables were retained in the exposure-outcome complete-case cohort.

```
analysis <- covid[
    covid$medcond_yn %in% c("Yes", "No") &
    covid$hosp_yn %in% c("Yes", "No"),
]

cat(sprintf("Rows: %s | Columns: %s\n",
    format(nrow(analysis), big.mark = ","),
    format(ncol(analysis), big.mark = ",")))
```

```
Rows: 1,464 | Columns: 12
```

The resulting exposure-outcome complete-case cohort contained 1464 cases (14.64% of the original sample). The remaining 8536 cases (85.36%) were excluded because exposure status, hospitalization status, or both were not recorded as Yes or No.

```
# Examine exposure and outcome reporting completeness
cat("Medical Condition Status (`medcond_yn`)\n")
```

```
Medical Condition Status (`medcond_yn`)
```

```
print(table(covid$medcond_yn, useNA = "ifany"))
```

```
Missing      No Unknown     Yes
   7358     973     826     843
```

```r
cat("Hospitalization Status (`hosp_yn`)\n")
```

```
Hospitalization Status (`hosp_yn`)
```

```r
print(table(covid$hosp_yn, useNA = "ifany"))
```

```
Missing      No Unknown     Yes
   3037    4720    1220    1023
```

```r
# Examine joint reporting patterns
cat("Joint Reporting Distribution\n")
```

```
Joint Reporting Distribution
```

```r
print(
    table(
        covid$medcond_yn,
        covid$hosp_yn,
        useNA = "ifany"
    )
)
```

```
          Missing   No Unknown  Yes
  Missing    2838 3397     470  653
  No          109  728     109   27
  Unknown      43  136     554   93
  Yes          47  459      87  250
```

```r
# Percentage with known medical-condition status
known_medical_condition <- mean(covid$medcond_yn %in% c("Yes", "No")) * 100

# Percentage with known hospitalization status
known_hospitalization <- mean(covid$hosp_yn %in% c("Yes", "No")) * 100

# Percentage with BOTH exposure and outcome known
known_both <- mean(
    covid$medcond_yn %in% c("Yes", "No") &
```

```
    covid$hosp_yn %in% c("Yes", "No")
) * 100

cat("Reporting Completeness (%)\n")
```

```
Reporting Completeness (%)
```

```
completeness_report <- data.frame(
    Metric = c(
        "Known Hospitalization Status",
        "Known Medical Condition Status",
        "Both Exposure and Outcome Known"
    ),
    Percentage = sprintf("%.2f%%", c(
        known_hospitalization,
        known_medical_condition,
        known_both
    ))
)

# Display reporting completeness
print(completeness_report, row.names = FALSE)
```

```
                         Metric Percentage
    Known Hospitalization Status     57.43%
  Known Medical Condition Status     18.16%
 Both Exposure and Outcome Known     14.64%
```

Reporting completeness differed substantially between variables. Hospitalization status was known for 57.43% of sampled records, while pre-existing medical condition status was known for only 18.16%. Both variables were known for 14.64% of sampled records.

Restricting the analysis to cases with known exposure and outcome status may introduce selection bias if the excluded cases differ systematically from the included cases.

3 Crude Association

3.1 Exposure-Outcome Table

We first estimate the crude association between the exposure (reported pre-existing medical condition) and outcome (reported hospitalization).

```r
# Construct 2x2 table with exposed/hospitalized first
exposure_outcome <- table(
    `Medical Condition (Exposure)` = factor(
        analysis$medcond_yn,
        levels = c("Yes", "No")
    ),
    `Hospitalized (Outcome)` = factor(
        analysis$hosp_yn,
        levels = c("Yes", "No")
    )
)

cat("Contingency Table\n")
```

```
Contingency Table
```

```r
# Display table with row and column totals
print(addmargins(
    exposure_outcome,
    FUN = list(Total = sum),
    quiet = TRUE
))
```

```
                           Hospitalized (Outcome)
Medical Condition (Exposure)  Yes   No Total
                       Yes    250  459   709
                       No      27  728   755
                       Total  277 1187  1464
```

3.2 Risk Ratio and Risk Difference

Here a and b represent hospitalized and non-hospitalized exposed cases, respectively. Similarly, c and d represent hospitalized and non-hospitalized unexposed cases, respectively. The crude risk ratio (RR) and risk difference (RD) are defined as:

$$RR = \frac{a/(a+b)}{c/(c+d)} \qquad RD = \frac{a}{a+b} - \frac{c}{c+d}$$

Hospitalization risk is estimated separately among exposed and unexposed cases.

```r
# Extract cells from exposure-outcome table
a <- exposure_outcome["Yes", "Yes"]
b <- exposure_outcome["Yes", "No"]
c <- exposure_outcome["No", "Yes"]
d <- exposure_outcome["No", "No"]

# Hospitalization risks
risk_exposed <- a / (a + b)
risk_unexposed <- c / (c + d)

# Effect measures
risk_ratio <- risk_exposed / risk_unexposed
risk_difference <- risk_exposed - risk_unexposed

cat("Risk Estimates & Measures of Effect\n")
```

```
Risk Estimates & Measures of Effect
```

```r
# Display hospitalization risks and effect measures
results <- data.frame(
    Measure = c(
        "Risk in Exposed (reported medical condition)",
        "Risk in Unexposed (no reported medical condition)",
        "Risk Ratio (RR)",
        "Risk Difference (RD)"
    ),
    Value = c(
        sprintf("%.3f (%.1f%%)", risk_exposed, risk_exposed * 100),
        sprintf("%.3f (%.1f%%)", risk_unexposed, risk_unexposed * 100),
        sprintf("%.3f", risk_ratio),
        sprintf("%.3f (%+.1f pp)", risk_difference, risk_difference * 100)
    )
)

print(results, row.names = FALSE)
```

```
                                          Measure            Value
      Risk in Exposed (reported medical condition)    0.353 (35.3%)
 Risk in Unexposed (no reported medical condition)     0.036 (3.6%)
                                  Risk Ratio (RR)            9.860
                              Risk Difference (RD) 0.317 (+31.7 pp)
```

Among cases in the complete-case analytic cohort, hospitalization was reported for 35.3% of individuals with a reported pre-existing medical condition and 3.6% of individuals without one. The crude risk ratio was 9.86, indicating that the proportion hospitalized was approximately 9.86 times as high among individuals with reported medical conditions. The crude risk difference was 31.7 percentage points, representing the absolute difference in hospitalization proportions between the two groups.

3.3 95% Confidence Interval for Risk Ratio

A 95% confidence interval (CI) for the crude risk ratio was calculated using the normal approximation to the log-tranformed risk ratio. The standard error of the log risk ratio is estimated as:

$$SE[\ln(RR)] = \sqrt{\frac{1}{a} - \frac{1}{a+b} + \frac{1}{c} - \frac{1}{c+d}}$$

The confidence interval is calculated on the logarithmic scale and then exponentiated to return to the risk-ratio scale.

$$95\%CI_{RR} = \exp[\ln(RR) \pm 1.96 \times SE[\ln(RR)]]$$

```
# Standard error of the log risk ratio
se_log_rr <- sqrt(
    (1 / a) - (1 / (a + b)) +
    (1 / c) - (1 / (c + d))
)

# Calculate confidence limits on log scale
log_rr <- log(risk_ratio)

log_rr_lower <- log_rr - (1.96 * se_log_rr)
log_rr_upper <- log_rr + (1.96 * se_log_rr)

# Transform confidence limits back to risk-ratio scale
rr_lower <- exp(log_rr_lower)
rr_upper <- exp(log_rr_upper)

cat("Risk Ratio (95% Confidence Interval)\n")
```

```
Risk Ratio (95% Confidence Interval)
```

```r
rr_summary <- data.frame(
    Estimate = sprintf("%.3f", risk_ratio),
    `95% CI Lower` = sprintf("%.3f", rr_lower),
    `95% CI Upper` = sprintf("%.3f", rr_upper),
    check.names = FALSE
)

print(rr_summary, row.names = FALSE)
```

```
 Estimate 95% CI Lower 95% CI Upper
    9.860        6.719       14.470
```

The crude risk ratio was 9.86 (95% CI: 6.72–14.47). The confidence interval excludes the null value of 1, indicating evidence of a positive crude association between reported pre-existing medical condition and hospitalization.

3.4 95% Confidence Interval for Risk Difference

A 95% confidence interval for the crude risk difference was calculated using the normal approximation for the difference between two independent proportions. The standard error is estimated as:

$$SE(RD) = \sqrt{\frac{R_1(1-R_1)}{n_1} + \frac{R_0(1-R_0)}{n_0}}$$

Here, R_1 and R_0 represent the observed hospitalization proportions among exposed and unexposed cases, respectively, while n_1 and n_0 represent the corresponding group sizes.

The confidence interval is calculated directly on the risk-difference scale:

$$95\%CI_{RD} = RD \pm 1.96 \times SE(RD)$$

```r
# Group sample sizes
n_exposed <- a + b
n_unexposed <- c + d

# Standard error of risk difference
se_rd <- sqrt(
    (risk_exposed * (1 - risk_exposed) / n_exposed) +
    (risk_unexposed * (1 - risk_unexposed) / n_unexposed)
```

```r
)

# Calculate confidence limits
rd_lower <- risk_difference - (1.96 * se_rd)
rd_upper <- risk_difference + (1.96 * se_rd)

cat("Risk Difference (95% Confidence Interval)\n")
```

```
Risk Difference (95% Confidence Interval)
```

```r
rd_summary <- data.frame(
  Scale = c("Proportion", "Percentage Points (pp)"),
  Estimate = c(
    sprintf("%+.3f", risk_difference),
    sprintf("%+.2f pp", risk_difference * 100)
  ),
  `95% CI Lower` = c(
    sprintf("%+.3f", rd_lower),
    sprintf("%+.2f pp", rd_lower * 100)
  ),
  `95% CI Upper` = c(
    sprintf("%+.3f", rd_upper),
    sprintf("%+.2f pp", rd_upper * 100)
  ),
  check.names = FALSE
)

print(rd_summary, row.names = FALSE)
```

```
                Scale  Estimate 95% CI Lower 95% CI Upper
           Proportion    +0.317       +0.279       +0.354
 Percentage Points (pp) +31.68 pp    +27.93 pp    +35.44 pp
```

The crude risk difference was 31.7 percentage points (95% CI: 27.9–35.4 percentage points). The confidence interval excludes the null value of 0, indicating evidence of a positive crude difference in hospitalization proportions between reported pre-existing medical condition and hospitalization.

These estimates describe crude, unadjusted associations and do not account for potential confounding by age or sex. Both confidence intervals were calculated using the normal approximation, which may be inaccurate for small sample sizes or extreme proportions.

4 Descriptive Analysis

Age and sex were hypothesized to be potential confounders based on the directed acyclic graph (DAG) and prior literature review. Before fitting adjusted models, their distributions were examined across exposure groups, along with the observed proportion hospitalized within each age and sex category.

4.1 Age

4.1.1 Age Distribution by Exposure

The age distribution was examined separately among cases with and without reported pre-existing medical conditions.

```
# Age distribution by exposure status
age_exposure <- table(
    `Age Group` = analysis$age_group,
    `Medical Condition` = analysis$medcond_yn
)

# Calculate column percentages (within exposure group)
age_exposure_prop <- prop.table(age_exposure, margin = 2) * 100

cat("Age Distribution by Exposure Status\n")
```

```
Age Distribution by Exposure Status
```

```
cat("1. Case Counts (with Margins)\n")
```

```
1. Case Counts (with Margins)
```

```
print(addmargins(
  age_exposure,
  margin = 1,
  FUN = list(Total = sum),
  quiet = TRUE
))
```

```
               Medical Condition
Age Group        No Yes
  0 - 9 Years   134  15
  10 - 19 Years  65  23
  20 - 29 Years 171  78
  30 - 39 Years 164  98
  40 - 49 Years  52  46
  50 - 59 Years 104  92
  60 - 69 Years  51 224
  70 - 79 Years   9  49
  80+ Years       5  84
  Total         755 709
```

```
cat("2. Column Percentages (% within exposure group)\n")
```

```
2. Column Percentages (% within exposure group)
```

```
# Combine count and % into a single clean string: "n (pct%)"
age_pct_formatted <- matrix(
  sprintf("%.1f%%", age_exposure_prop),
  nrow = nrow(age_exposure),
  dimnames = dimnames(age_exposure)
)

print(age_pct_formatted, quote = FALSE)
```

```
               Medical Condition
Age Group       No    Yes
  0 - 9 Years   17.7% 2.1%
  10 - 19 Years 8.6%  3.2%
  20 - 29 Years 22.6% 11.0%
  30 - 39 Years 21.7% 13.8%
  40 - 49 Years 6.9%  6.5%
  50 - 59 Years 13.8% 13.0%
  60 - 69 Years 6.8%  31.6%
  70 - 79 Years 1.2%  6.9%
  80+ Years     0.7%  11.8%
```

Column percentages represent the age distribution within each exposure group, $P(Age \mid E)$. Each exposure column sums to 100%.

```r
# Age groups aged 60+
older_groups <- c(
    "60 - 69 Years",
    "70 - 79 Years",
    "80+ Years"
)

# Case counts and totals
n_older <- colSums(age_exposure[older_groups, , drop = FALSE])
n_total <- colSums(age_exposure)

# Percentage  aged 60+ in each exposure group
older_percent <- (n_older / n_total) * 100

cat("Proportion Aged 60+ by Exposure Status\n")
```

```
Proportion Aged 60+ by Exposure Status
```

```r
older_summary <- data.frame(
  `Medical Condition` = names(older_percent),
  `Aged 60+ (n)` = format(n_older, big.mark = ","),
  `Total (N)` = format(n_total, big.mark = ","),
  `Percentage Aged 60+` = sprintf("%.1f%%", older_percent),
  check.names = FALSE
)

print(older_summary, row.names = FALSE)
```

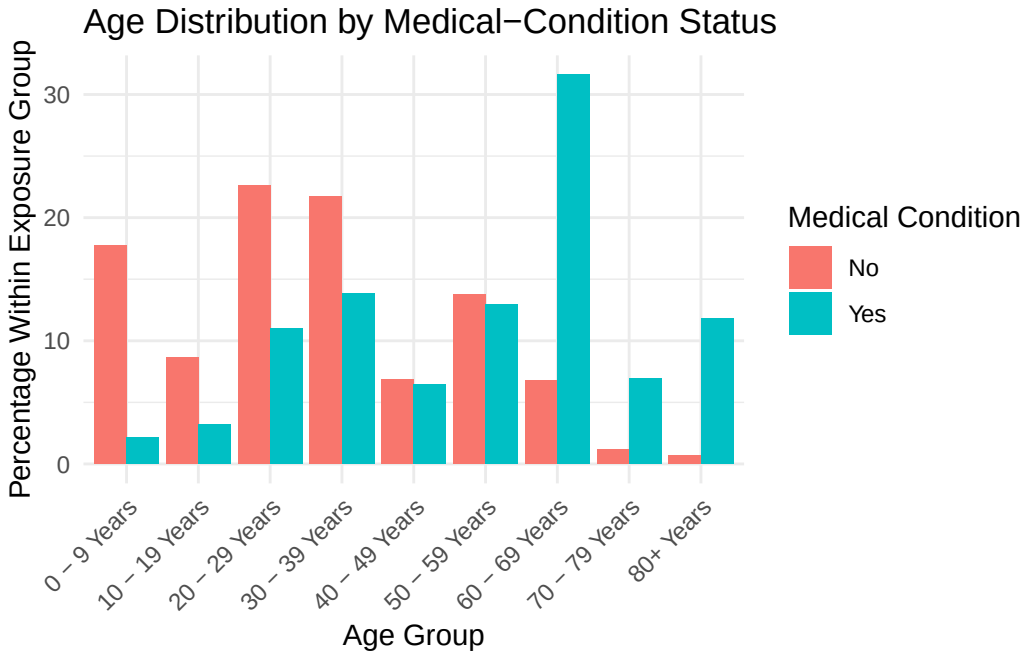
```
 Medical Condition Aged 60+ (n) Total (N) Percentage Aged 60+
                No           65       755                8.6%
               Yes          357       709               50.4%
```

The age distribution differed substantially between exposure groups. Among cases with reported pre-existing medical conditions, 50.4% were aged 60 years or older, compared with 8.6% among cases without reported conditions. Older individuals were disproportionately represented in the exposed group, suggesting that age may contribute to the observed crude association between reported pre-existing medical condition and hospitalization.

```r
# Convert column percentages to a data frame for plotting
age_exposure_df <- as.data.frame(
    prop.table(age_exposure, margin = 2) * 100
)

names(age_exposure_df) <- c(
    "Age",
    "Medical_Condition",
    "Percentage"
)

# Plot age distribution within each exposure group
ggplot(
    age_exposure_df,
    aes(
        x = Age,
        y = Percentage,
        fill = Medical_Condition
    )
) +
    geom_col(position = "dodge") +
    labs(
        title = "Age Distribution by Medical-Condition Status",
        x = "Age Group",
        y = "Percentage Within Exposure Group",
        fill = "Medical Condition"
    ) +
    theme_minimal() +
    theme(
        axis.text.x = element_text(angle = 45, hjust = 1)
    )
```



4.1.2 Hospitalization by Age

The proportion hospitalized was calculated within each age group to examine whether observed hospitalization proportion varies by age.

```
# Hospitalization status by age group
age_outcome <- table(
    `Age Group` = analysis$age_group,
    `Hospitalized` = analysis$hosp_yn
)

# Row percentages: hospitalization status within each age group
age_outcome_pct <- prop.table(age_outcome, margin = 1) * 100

cat("Hosptialization Distribution by Age Group\n")
```

```
Hosptialization Distribution by Age Group
```

```
cat("1. Case Counts (with Row Margins)\n")
```

```
1. Case Counts (with Row Margins)
```

```r
print(addmargins(
  age_outcome,
  margin = 2,
  FUN = list(Total = sum),
  quiet = TRUE
))
```

```
              Hospitalized
Age Group        No Yes Total
  0 - 9 Years   145   4   149
  10 - 19 Years  86   2    88
  20 - 29 Years 240   9   249
  30 - 39 Years 238  24   262
  40 - 49 Years  87  11    98
  50 - 59 Years 184  12   196
  60 - 69 Years 127 148   275
  70 - 79 Years  27  31    58
  80+ Years      53  36    89
```

```r
cat("\n2. Row Percentages (% within age group, summing across rows)\n")
```

```
2. Row Percentages (% within age group, summing across rows)
```

```r
age_pct_formatted <- matrix(
  sprintf("%.1f%%", age_outcome_pct),
  nrow = nrow(age_outcome),
  dimnames = dimnames(age_outcome)
)
print(age_pct_formatted, quote = FALSE)
```

```
              Hospitalized
Age Group       No    Yes
  0 - 9 Years   97.3% 2.7%
  10 - 19 Years 97.7% 2.3%
  20 - 29 Years 96.4% 3.6%
  30 - 39 Years 90.8% 9.2%
  40 - 49 Years 88.8% 11.2%
  50 - 59 Years 93.9% 6.1%
  60 - 69 Years 46.2% 53.8%
```

```
  70 - 79 Years 46.6% 53.4%
  80+ Years     59.6% 40.4%
```

Row percentages represent hospitalization status within each age group, $P(Y \mid Age)$. Each age-group row sums to 100%.

```
# Calculate percentage hospitalized within each age group
hosp_yes_count <- age_outcome[, "Yes"]
total_in_age <- rowSums(age_outcome)
age_hosp_percent <- (hosp_yes_count / total_in_age) * 100

cat("Risk of Hospitalization by Age Group\n")
```

```
Risk of Hospitalization by Age Group
```

```
hosp_risk_summary <- data.frame(
  `Age Group` = names(age_hosp_percent),
  `Hospitalized (n)` = format(hosp_yes_count, big.mark = ","),
  `Total Cases (N)` = format(total_in_age, big.mark = ","),
  `Hospitalization Risk` = sprintf("%.1f%%", age_hosp_percent),
  check.names = FALSE
)

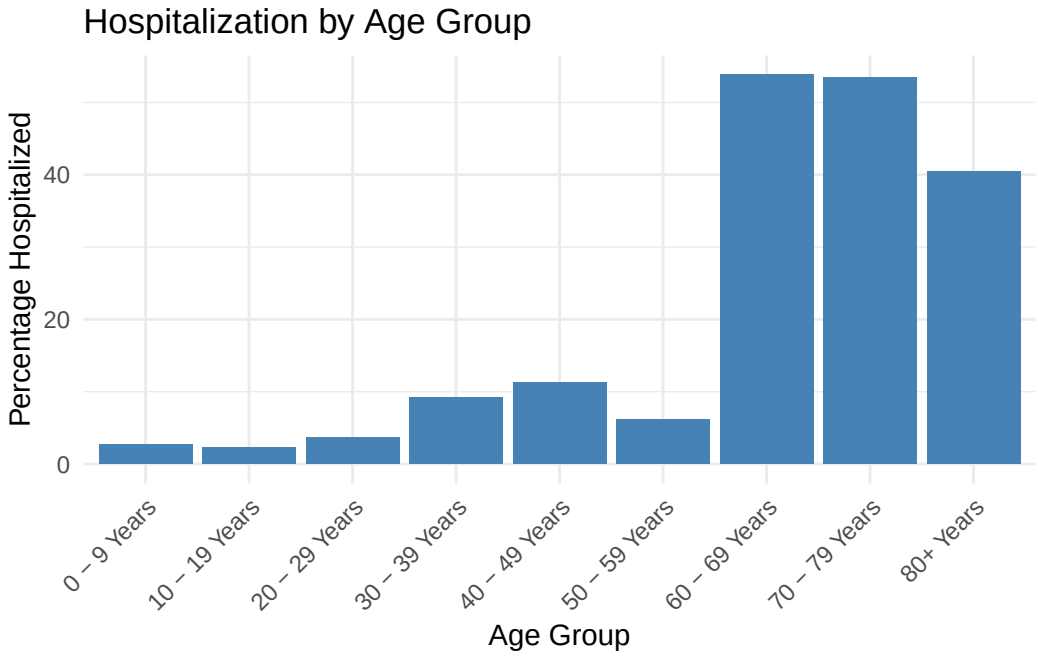
print(hosp_risk_summary, row.names = FALSE)
```

```
     Age Group Hospitalized (n) Total Cases (N) Hospitalization Risk
   0 - 9 Years                4             149                 2.7%
 10 - 19 Years                2              88                 2.3%
 20 - 29 Years                9             249                 3.6%
 30 - 39 Years               24             262                 9.2%
 40 - 49 Years               11              98                11.2%
 50 - 59 Years               12             196                 6.1%
 60 - 69 Years              148             275                53.8%
 70 - 79 Years               31              58                53.4%
     80+ Years               36              89                40.4%
```

Hospitalization proportions varied substantially across age groups. Among cases aged 60–69 years, 53.8% were hospitalized, compared with 3.6% among cases aged 20–29 years. The proportion hospitalized among cases aged 80 years or older was 40.4%, indicating that the observed pattern did not increase consistently across all age categories. These findings suggest a strong descriptive association between age and hospitalization within the analytic cohort.

```r
# Convert hospitalization percentages into a data frame for plotting
age_hosp_df <- data.frame(
    Age = names(age_hosp_percent),
    Percentage = as.numeric(age_hosp_percent)
)

# Plot hospitalization percentage within each age group
ggplot(
    age_hosp_df,
    aes(
        x = Age,
        y = Percentage
    )
) +
    geom_col(fill = "steelblue") +
    labs(
        title = "Hospitalization by Age Group",
        x = "Age Group",
        y = "Percentage Hospitalized"
    ) +
    theme_minimal() +
    theme(
        axis.text.x = element_text(angle = 45, hjust = 1)
    )
```



4.2 Sex

4.2.1 Sex by Exposure

The sex distribution was examined separately among cases with and without reported pre-existing medical conditions.

```
# Sex distribution by exposure status
sex_exposure <- table(
    analysis$sex,
    analysis$medcond_yn
)

# Counts
sex_exposure
```

```
           No Yes
  Female  368 327
  Male    386 381
  Missing   1   0
  Unknown   0   1
```

```r
# Column percentages: sex distribution within each exposure group
round(
    prop.table(sex_exposure, margin = 2) * 100,
    1
)
```

```
           No  Yes
  Female  48.7 46.1
  Male    51.1 53.7
  Missing  0.1  0.0
  Unknown  0.0  0.1
```

Column percentages represent the sex distribution within each exposure group, $P(Sex \mid E)$. Each exposure column sums to 100%.

```r
# Calculate percentage of males within each exposure group
male_percent <- prop.table(
    sex_exposure,
    margin = 2
)["Male", ] * 100

round(male_percent, 1)
```

```
  No  Yes
51.1 53.7
```

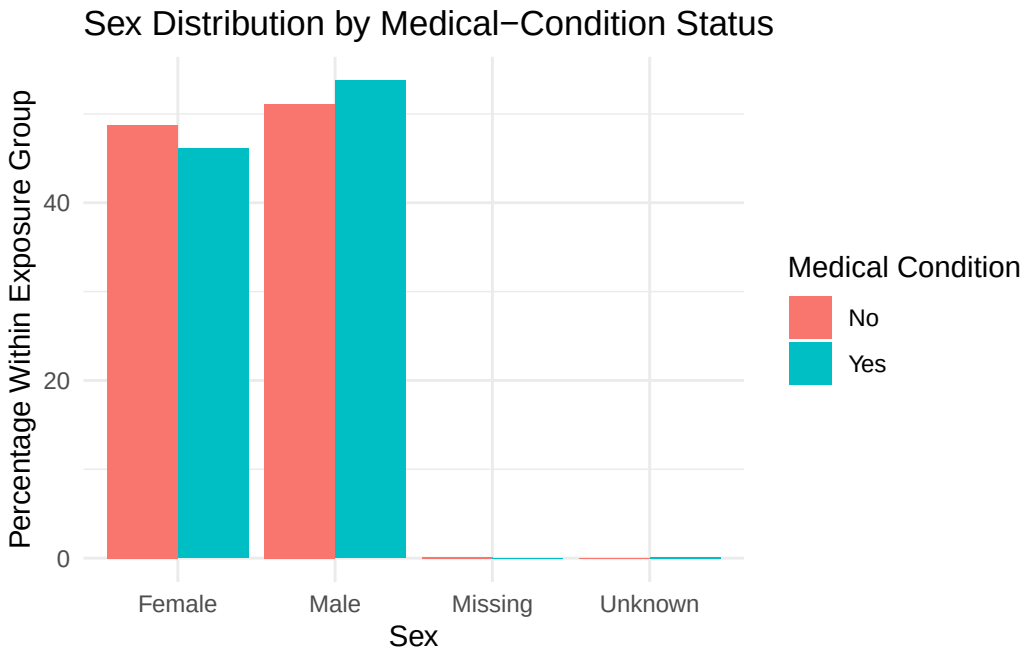
The sex distribution was similar between exposure groups. Males accounted for 53.7% of cases with reported pre-existing medical conditions, compared with 51.1% of cases without reported conditions. These relatively small differences suggest that sex may contribute less to the crude exposure-outcome association than age, although this descriptive comparison alone does not establish the extent of confounding.

```r
# Convert column percentages to a data frame for plotting
sex_exposure_df <- as.data.frame(
    prop.table(sex_exposure, margin = 2) * 100
)

names(sex_exposure_df) <- c(
    "Sex",
```

```
    "Medical_Condition",
    "Percentage"
)

# Plot sex distribution within each exposure group
ggplot(
    sex_exposure_df,
    aes(
        x = Sex,
        y = Percentage,
        fill = Medical_Condition
    )
) +
    geom_col(position = "dodge") +
    labs(
        title = "Sex Distribution by Medical-Condition Status",
        x = "Sex",
        y = "Percentage Within Exposure Group",
        fill = "Medical Condition"
    ) +
    theme_minimal()
```



4.2.2 Hospitalization by Sex

The proportion hospitalized was calculated within each sex category to examine differences in hospitalization risk between males and females.

```
# Hospitalization status by sex
sex_outcome <- table(
    analysis$sex,
    analysis$hosp_yn
)

# Counts
sex_outcome
```

```
           No Yes
  Female  573 122
  Male    612 155
  Missing   1   0
  Unknown   1   0
```

```
# Row percentages: hospitalization status within each sex group
round(
    prop.table(sex_outcome, margin = 1) * 100,
    1
)
```

```
             No   Yes
  Female   82.4  17.6
  Male     79.8  20.2
  Missing 100.0   0.0
  Unknown 100.0   0.0
```

Row percentages represent hospitalization status within each sex category, $P(Y \mid Sex)$. Each sex row sums to 100%.

```
# Calculate percentage hospitalized within each sex category
sex_hosp_percent <- prop.table(
    sex_outcome,
    margin = 1
```

```
)[, "Yes"] * 100

round(sex_hosp_percent, 1)
```

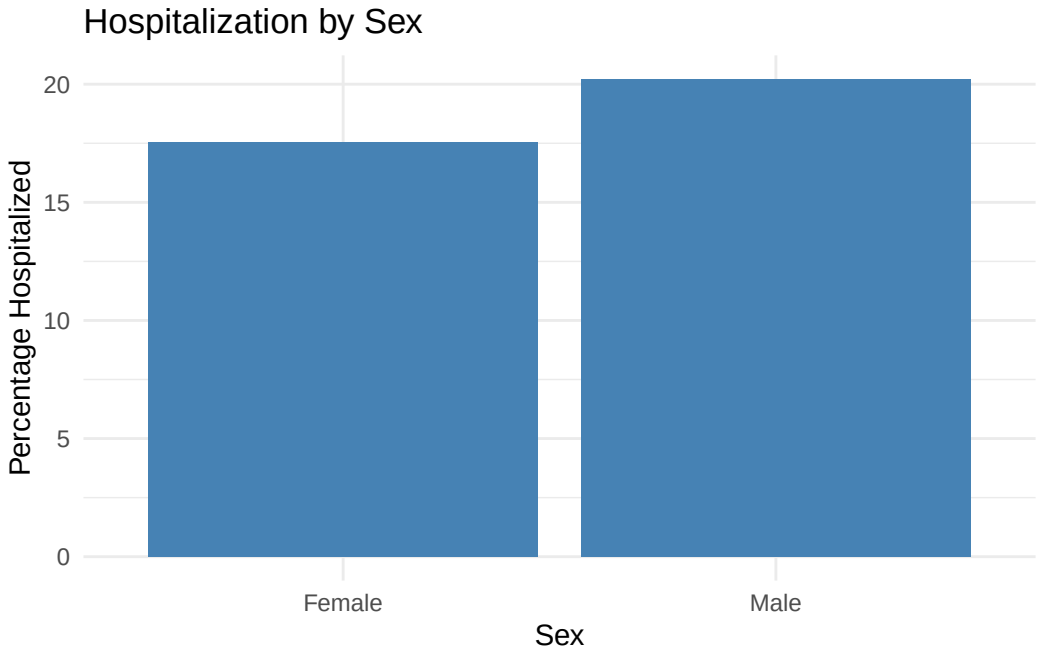
```
 Female    Male Missing Unknown
   17.6    20.2     0.0     0.0
```

Hospitalization proportions differed modestly between males and females. Among male cases, 20.2% were hospitalized, compared with 17.6% among female cases. These results suggest a relatively small descriptive association between sex and hospitalization within the analytic cohort.

```
# Convert hospitalization percentages into a data frame for plotting
sex_hosp_df <- data.frame(
    Sex = names(sex_hosp_percent),
    Percentage = as.numeric(sex_hosp_percent)
)

# Keep Female and Male categories for visualization
sex_hosp_df <- subset(
    sex_hosp_df,
    Sex %in% c("Female", "Male")
)

# Plot hospitalization percentage within each sex category
ggplot(
    sex_hosp_df,
    aes(
        x = Sex,
        y = Percentage
    )
) +
    geom_col(fill = "steelblue") +
    labs(
        title = "Hospitalization by Sex",
        x = "Sex",
        y = "Percentage Hospitalized"
    ) +
    theme_minimal()
```



4.3 Summary of Potential Confounders

The descriptive comparisons were used to examine the relationships of age and sex with both reported pre-existing medical conditions and hospitalization. Differences in these distributions may contribute to the observed crude association.

Age and sex were retained as adjustment variables based on the hypothesized causal structure and prior literature review rather than solely on the descriptive results. The following adjusted analysis examines the exposure-outcome association while accounting for these variables.

5 Adjusted Analysis

Age and sex were identified as potential confounders based on the hypothesized causal structure and prior literature review. To examine the association between reported pre-existing medical conditions and hospitalization while accounting for these variables, a multivariable logistic regression model was fitted.

5.1 Adjusted Analytic Cohort

The adjusted analysis was restricted to cases with known exposure, outcome, age, and sex status. Two records in the exposure-outcome complete-case cohort had sex coded as `Missing` or `Unknown` and have been excluded.

```r
# Restrict analysis to cases with recorded Female or Male sex
adjusted_analysis <- analysis[
    analysis$sex %in% c("Female", "Male") &
    !is.na(analysis$age_group) &
    analysis$age_group != "Missing" &
    analysis$age_group != "Unknown",
]

# Display adjusted cohort size
nrow(adjusted_analysis)
```

```
[1] 1462
```

The resulting adjusted-analysis cohort contained 1462 cases.

The outcome, exposure, and adjustment variables were encoded as categorical factors to define their reference categories explicitly.

```r
# Hospitalization: No is the reference, Yes is the event
adjusted_analysis$hospitalized <- factor(
    adjusted_analysis$hosp_yn,
    levels = c("No", "Yes")
)

# Medical condition: No is the reference
adjusted_analysis$medcond <- factor(
    adjusted_analysis$medcond_yn,
    levels = c("No", "Yes")
)

# Sex: Female is the reference
adjusted_analysis$sex_factor <- factor(
    adjusted_analysis$sex,
    levels = c("Female", "Male")
)

# Age: 0-9 years is the reference
adjusted_analysis$age_factor <- factor(
    adjusted_analysis$age_group,
    levels = c(
        "0 - 9 Years",
        "10 - 19 Years",
```

```
        "20 - 29 Years",
        "30 - 39 Years",
        "40 - 49 Years",
        "50 - 59 Years",
        "60 - 69 Years",
        "70 - 79 Years",
        "80+ Years"
    )
)

# Verify reference categories
lapply(
    adjusted_analysis[
        c("hospitalized", "medcond", "age_factor", "sex_factor")
    ],
    levels
)
```

```
$hospitalized
[1] "No"  "Yes"

$medcond
[1] "No"  "Yes"

$age_factor
[1] "0 - 9 Years"   "10 - 19 Years" "20 - 29 Years" "30 - 39 Years"
[5] "40 - 49 Years" "50 - 59 Years" "60 - 69 Years" "70 - 79 Years"
[9] "80+ Years"

$sex_factor
[1] "Female" "Male"
```

```
# Verify that all model variables are complete
colSums(is.na(
    adjusted_analysis[
        c("hospitalized", "medcond", "age_factor", "sex_factor")
    ]
))
```

```
hospitalized      medcond   age_factor   sex_factor
           0            0            0            0
```

Hospitalization was modeled with `Yes` as the event and `No` as the reference category. Reported pre-existing medical-condition status used `No` as the reference category. Sex used `Female` as the reference category, while age used `0-9 Years` as the reference group.

5.2 Logistic Regression

A multivariable logistic regression model was fit with hospitalization as the binary outcome, reported pre-existing medical condition as the exposure, and age group and sex as adjustment variables.

```
# Fit logistic regression adjusting for age and sex
logistic_model <- glm(
    hospitalized ~ medcond + age_factor + sex_factor,
    data = adjusted_analysis,
    family = binomial(link = "logit")
)

# Examine model coefficients and standard errors
summary(logistic_model)
```

```
Call:
glm(formula = hospitalized ~ medcond + age_factor + sex_factor,
    family = binomial(link = "logit"), data = adjusted_analysis)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)             -4.180e+00  5.354e-01  -7.807 5.86e-15 ***
medcondYes               1.874e+00  2.319e-01   8.079 6.50e-16 ***
age_factor10 - 19 Years -5.267e-01  8.929e-01  -0.590 0.555311
age_factor20 - 29 Years -1.615e-01  6.273e-01  -0.258 0.796792
age_factor30 - 39 Years  6.146e-01  5.667e-01   1.084 0.278178
age_factor40 - 49 Years  7.327e-01  6.196e-01   1.183 0.237001
age_factor50 - 59 Years  2.942e-05  6.066e-01   0.000 0.999961
age_factor60 - 69 Years  2.624e+00  5.424e-01   4.838 1.31e-06 ***
age_factor70 - 79 Years  2.567e+00  5.977e-01   4.295 1.75e-05 ***
age_factor80+ Years      1.964e+00  5.850e-01   3.358 0.000785 ***
sex_factorMale           2.805e-01  1.824e-01   1.538 0.123996
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1419.45  on 1461  degrees of freedom
Residual deviance:  947.29  on 1451  degrees of freedom
AIC: 969.29

Number of Fisher Scoring iterations: 6
```

Logistic regression estimates associations on the log-odds scale. As a result, the exposure coefficient must be exponentiated to obtain an adjusted odds ratio.

```r
# Extract the exposure coefficient (log odds ratio)
exposure_beta <- coef(logistic_model)["medcondYes"]

# Convert the coefficient to an adjusted odds ratio
adjusted_or <- exp(exposure_beta)

# Display adjusted odds ratio
round(adjusted_or, 3)
```

```
medcondYes
     6.513
```

A 95% confidence interval for the adjusted odds ratio was calculated using the standard error of the exposure coefficient. Confidence limits were calculated on the log-odds scale and exponentiated to return to the odds-ratio scale.

$$\exp[\hat{\beta}_E \pm 1.96 \times SE(\hat{\beta}_E)]$$

5.2.1 Interpretation of Logistic Regression

Logistic regression models the log odds of a binary outcome as a function of the exposure and adjustment varialbes. The fitted model can be written as:

$$\beta_0 + \beta_E E_i + \sum_{k=1}^{8} \beta_{A_k} A_{ik} + \beta_S S_i,$$

where

$$p_i = P(Y_i = 1 \mid E_i, A_i, S_i).$$

Here, E_i represents reported pre-existing medical condition status, A_{ik} represents the eight indicator variables from the non-reference age groups, and S_i represents sex.

The exposure coefficient β_E represents the difference in conditional log odds of hospitalization between cases with and without reported pre-existing medical conditions, holding age group and sex constant.

5.3 Adjusted Odds Ratio

A 95% Wald confidence interval was calculated on the log-odds scale using the estimated coefficient and its standard error, then exponentiated to obtain confidence limits on the odds-ratio scale.

$$OR_{\text{adjusted}} = e^{\hat{\beta}_E}$$

$$95\%CI_{OR} = \exp(\hat{\beta}_E \pm 1.96 \times SE(\hat{\beta}_E))$$

```
# Extract the exposure coefficient (log odds ratio)
beta <- coef(logistic_model)["medcondYes"]

# Exponentiate to obtain the adjusted odds ratio
adjusted_or <- exp(beta)

# Extract the coefficient variance and calculate its standard error
se <- sqrt(vcov(logistic_model)["medcondYes", "medcondYes"])

# Calculate 95% Wald confidence interval on the odds-ratio scale
or_ci <- exp(beta + c(-1, 1) * 1.96 * se)

# Display adjusted odds ratio and confidence interval
round(
    c(
        Adjusted_OR = adjusted_or,
        Lower_95_CI = or_ci[1],
        Upper_95_CI = or_ci[2]
    ),
    3
)
```

```
Adjusted_OR.medcondYes           Lower_95_CI           Upper_95_CI
                 6.513                 4.134                10.260
```

The fitted model estimate an adjusted odds ratio of 6.51 (95% CI: 4.13–10.26) for reported pre-existing medical conditions, conditional on age group and sex. This indicates that the estimated odds of hospitalization were higher among cases with reported pre-existing medical conditions than among cases without reported conditions, after accounting for age and sex.

5.4 Regression Standardization

Regression standardization was used to estimate hospitalization proportions for hte two exposure conditions while accounting for the age and sex distribution of the adjusted-analysis cohort. The fitted logistic regression model was used to predict hospitalization probabilities after setting medical-condition status to Yes for all cases and, separately, to No for all cases, while retaining each case's observed age and sex. Predictions were then averaged across the same cohort in both scenarios.

5.4.1 Constructing Exposure Scenarios

Two copies of the adjusted-analysis cohort were created. Pre-existing medical condition status was set to `Yes` for every case in one copy and `No` for every case in the other copy. Age and sex were left unchanged in both copies.

```
# Create two copies of the adjusted cohort
everyone_exposed <- adjusted_analysis
everyone_unexposed <- adjusted_analysis

# Assign reported medical conditions to every case
everyone_exposed$medcond <- factor(
    "Yes",
    levels = levels(adjusted_analysis$medcond)
)

# Assign no reported medical conditions to every case
everyone_unexposed$medcond <- factor(
    "No",
    levels = levels(adjusted_analysis$medcond)
)
```

These scenarios are used to generate model-based predictions. They do not represent actual observed data, but rather hypothetical situations in which every case is assigned the same exposure status while retaining their observed age and sex.

5.4.2 Predicting Standardized Hospitalization Proportions

The fitted logistic regression model was used to predict hospitalization probabilities for each case under both exposure scenarios.

Setting `type = "response"` returns the predicted probabilities rather than predicted log odds

```
# Predict hospitalization probabilities under each exposure scenario
pred_exposed <- predict(
    logistic_model,
    newdata = everyone_exposed,
    type = "response"
)

pred_unexposed <- predict(
    logistic_model,
    newdata = everyone_unexposed,
    type = "response"
)

# Average predicted probabilities across the adjusted cohort
adjusted_risk_exposed <- mean(pred_exposed)
adjusted_risk_unexposed <- mean(pred_unexposed)

# Display standardized hospitalization proportions
round(
    c(
        Exposed = adjusted_risk_exposed,
        Unexposed = adjusted_risk_unexposed
    ),
    4
)
```

```
  Exposed Unexposed
   0.2551    0.0669
```

The standardized hospitalization proportion under reported pre-existing medical conditions was 25.5%, compared with 6.7% under no reported medical conditions.

These are model-based estimates averaged across the observed age and sex distribution of the adjusted-analysis cohort, rather than the directly observed hospitalization proportions within each exposure group.

5.5 Adjusted Risk Ratio and Risk Difference

The standardized hospitalization proportions were used to calculate an adjusted risk ratio and risk difference.

The adjusted risk ratio compares the two standardized proportions:

$$RR_{\text{adjusted}} = \frac{\bar{p}E = 1}{\bar{p}E = 0}$$

The adjusted risk difference represents their absolute difference:

$$RD_{\text{adjusted}} = \bar{p}E = 1 - \bar{p}E = 0$$

Where $\bar{p}E = 1$ and $\bar{p}E = 0$ are the mean model-predicted hospitalizaiton probabilities under the exposed and unexposed scenarios, respectively.

```
# Calculate adjusted risk ratio and risk difference
adjusted_rr <- adjusted_risk_exposed / adjusted_risk_unexposed
adjusted_rd <- adjusted_risk_exposed - adjusted_risk_unexposed

# Display adjusted association measures
round(
    c(
        Adjusted_RR = adjusted_rr,
        Adjusted_RD = adjusted_rd,
        Adjusted_RD_Percentage_Points = adjusted_rd * 100
    ),
    3
)
```

```
                  Adjusted_RR                   Adjusted_RD
                        3.816                         0.188
Adjusted_RD_Percentage_Points
                       18.825
```

The standardized adjusted risk ratio was 3.82, indicating that the model-estimated hospitalization proportion under reported pre-existing medical conditions was approximately 3.82 times the proportion under no reported medical conditions, after standardizing to the same age and sex distribution.

The adjusted risk difference was 18.8 percentage points, representing the absolute difference between the two standardized hospitalization proportions.

These estimates describe model-adjusted associations within the analytic cohort. They do not establish a causal effect of pre-existing medical conditions on hospitalization. In particular, adjustment for age and sex does not eliminate potential residual confounding, selection bias associated with missing surveillance data, or limitations arising from the sampling procedure.

6 Statistical Uncertainty and Bootstrap Inference

The nonparametric bootstrap approximates the sampling distribution of a statistic by repeatedly drawing samples with replacement from the observed data. The variability of the resulting estimates can be used to quantify sampling uncertainty.

6.1 Resampling the Analytic Cohort

Each bootstrap sample contains the same number of observations as the original analytic cohort. However, because sampling occurs with replacement, some observations appear multiple times while others are not selected.

Maintaining the original sample size allows the bootstrap procedure to approximate how an estimator would vary across repeated samples of that size, using the observed data as an empirical approximation to the underlying population distribution.

```
# Set random seed for reproducibility
set.seed(42)

# Number of observations in adjusted-analysis cohort
n <- nrow(adjusted_analysis)

# Draw n row indices with replacement
bootstrap_indices <- sample(
    x = seq_len(n),
    size = n,
    replace = TRUE
)

# Construct a bootstrap sample
bootstrap_sample <- adjusted_analysis[
    bootstrap_indices,
]

# Compare sample sizes
nrow(adjusted_analysis)
```

```
[1] 1462
```

```
nrow(bootstrap_sample)
```

```
[1] 1462
```

For a dataset containing n observations, the probability that a specific observation is not selected in a single draw is:

$$1 - \frac{1}{n}.$$

The probability that a specific observation is never selected across n independent draws is:

$$P(\text{not selected}) = (1 - \frac{1}{n})^n.$$

As n becomes large, this approaches $e^{-1} \approx 0.368$. As a result, the probability that an observation appears at least once is roughly $1 - e^{-1} \approx 0.632$.

```
# Unique observations selected
n_unique <- length(unique(bootstrap_indices))

# Proportion of original observations represented
prop_unique <- n_unique / n

n_unique
```

```
[1] 949
```

```
prop_unique
```

```
[1] 0.6491108
```

```
# Count how many times each original observation appears
selection_counts <- tabulate(
    bootstrap_indices,
    nbins = n
)

# Frequency distribution of selection counts
table(selection_counts)
```

```
selection_counts
  0   1   2   3   4   5
513 564 277  90  16   2
```

```
# Original hospitalization distribution
prop.table(table(adjusted_analysis$hospitalized))
```

```
       No       Yes
0.8105335 0.1894665
```

```
# Bootstrap hospitalization distribution
prop.table(table(bootstrap_sample$hospitalized))
```

```
       No       Yes
0.8187415 0.1812585
```

6.1.1 Constructing a Bootstrap Distribution

A single bootstrap sample demonstrates how sampling with replacement changes the composition of the analytic cohort. However, estimating sampling variability requires repeated resampling and recalculation of the statistic of interest for each bootstrap sample.

```
# Examine hospitalization variable
class(adjusted_analysis$hospitalized)
```

```
[1] "factor"
```

```
levels(adjusted_analysis$hospitalized)
```

```
[1] "No"  "Yes"
```

```
table(adjusted_analysis$hospitalized)
```

```
  No  Yes
1185  277
```

```r
# Verify the proportion
mean(adjusted_analysis$hospitalized == "Yes")
```

```
[1] 0.1894665
```

We first examine the overall hospitalization proportion. The observed hospitalization proportion is defined as:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^{n} Y_i,$$

where $Y_i = 1$ indicates hospitalization for case i, $Y_i = 0$ indicates no hospitalization, and n is the number of cases in the analytic cohort.

For each boostrap replication b, we draw n case records with replacement and recalculate the hospitalization proportion, producing $\hat{p}^{*(b)}$. Repeating this process B times yields an empirical bootstrap distribution of the estimator.

```r
# Original hospitalization proportion
original_prop <- mean(adjusted_analysis$hospitalized == "Yes")

original_prop
```

```
[1] 0.1894665
```

```r
set.seed(42)

# Number of bootstrap replications
B <- 1000

# Store hospitalization proportions
bootstrap_props <- numeric(B)

# Repeat the resampling process
for (b in seq_len(B)) {

    # Sample observations with replacement
    indices <- sample(
        x = seq_len(n),
        size = n,
```

```
        replace = TRUE
    )

    # Construct bootstrap dataset
    boot_data <- adjusted_analysis[indices, ]

    # Calculate hospitalization proportion
    bootstrap_props[b] <- mean(
        boot_data$hospitalized == "Yes"
    )
}

# Examine bootstrap estimates
summary(bootstrap_props)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.1587  0.1826  0.1895  0.1895  0.1956  0.2257
```

The bootstrap estimates provide an empirical approximation to the sampling distribution of the hospitalization proportion. Examining this distribution allows us to evaluate its center, spread, and shape.

The distribution's center indicates whether the bootstrap estimates tend to cluster around the original observed proportion. Its spread reflects the variability of the estimator across resamples, while its shape helps assess whether a symmetric, approximately normal approximation is reasonable.

The histogram below displays the 1,000 bootstrap estimates. The solid red line indicates the original hospitalization proportion, while the dashed green line indicates the mean of the bootstrap estimates.

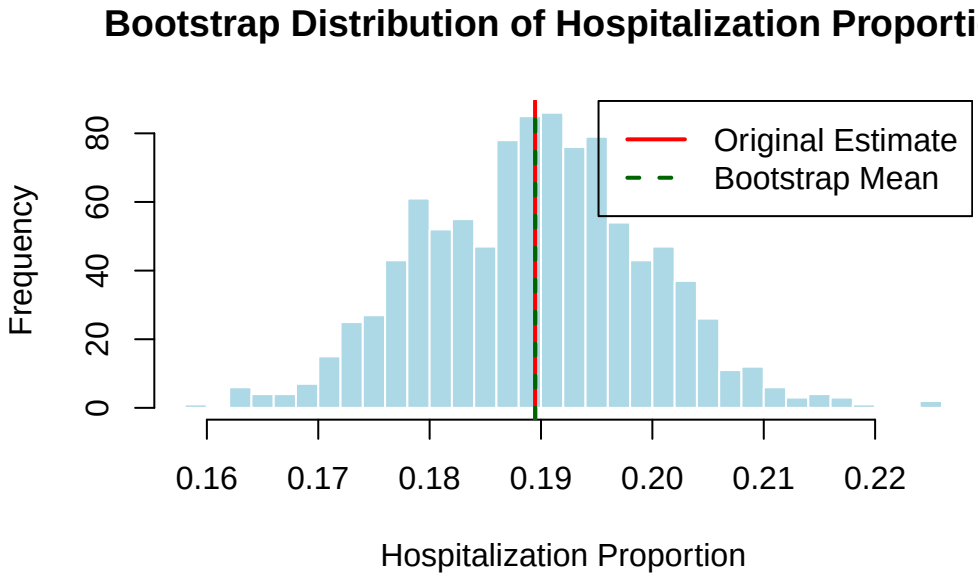
```
hist(
    bootstrap_props,
    breaks = 30,
    main = "Bootstrap Distribution of Hospitalization Proportion",
    xlab = "Hospitalization Proportion",
    col = "lightblue",
    border = "white"
)

# Original observed proportion
abline(
```

```
    v = original_prop,
    col = "red",
    lwd = 2
)

# Bootstrap mean
abline(
    v = mean(bootstrap_props),
    col = "darkgreen",
    lwd = 2,
    lty = 2
)

legend(
    "topright",
    legend = c("Original Estimate", "Bootstrap Mean"),
    col = c("red", "darkgreen"),
    lty = c(1, 2),
    lwd = 2
)
```



The bootstrap distribution appears approximately symmetric, with estimates concentrated around 0.189, compared with the original observed hospitalization proportion of 0.189. The proximity of these values suggests that the boostrap estimates are centered near the observed statistic

The bootstrap distribution demonstrates that the estimated hospitalization proportion varies across samples of the same size, even when all samples are drawn from the same observed cohort.

The bootstrap standard error is calculated as the sample standard deviation of the B bootstrap estimates:

$$\widehat{\mathrm{SE}}_{\mathrm{boot}}(\hat{p}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}(\hat{p}^{*(b)} - \bar{p}^{*})^2}.$$

Here, $\hat{p}^{*(b)}$ represents the estimated hospitalization proportion from bootstrap replication b, and $\bar{p}^{*}$ is the mean of the bootstrap estimates.

Unlike the standard deviation of individual hospitalization outcomes, which describes variation across cases, the bootstrap standard error describes variation in the estimated hospitalization proportion across resamples.

We can also construct a 95% percentile bootstrap confidence interval by identifying the 2.5th and 97.5th percentiles of the bootstrap distribution:

$$CI_{95\%} = [Q_{0.025}(\hat{p}^{*}), Q_{0.975}(\hat{p}^{*})].$$

This approach uses the empirical distribution of the bootstrap estimates rather than assuming that the distribution is exactly normal.

```
# Calculate bootstrap summary statistics
bootstrap_ci <- quantile(
    bootstrap_props,
    probs = c(0.025, 0.975)
)

bootstrap_summary <- data.frame(
    Statistic = c(
        "Original hospitalization proportion",
        "Mean bootstrap proportion",
        "Bootstrap standard error",
        "95% CI lower bound",
        "95% CI upper bound"
    ),
    Estimate = c(
        original_prop,
        mean(bootstrap_props),
        sd(bootstrap_props),
```

```
        bootstrap_ci[1],
        bootstrap_ci[2]
    )
)

knitr::kable(
    bootstrap_summary,
    digits = 4,
    caption = "Bootstrap inference for the hospitalization proportion"
)
```

Table 1: Bootstrap inference for the hospitalization proportion

Statistic	Estimate
Original hospitalization proportion	0.1895
Mean bootstrap proportion	0.1895
Bootstrap standard error	0.0101
95% CI lower bound	0.1703
95% CI upper bound	0.2086

The observed hospitalization proportion in the adjusted analytic cohort was 18.9466484%. Across 1000 bootstrap replications, the mean estimated hospitalization proportion was also approximately 18.9487688%, with a bootstrap standard error of 0.0100507 percentage points.

The 95% percentile bootstrap confidence interval ranged from `rbootstrap_ci[1] * 100`% to `rbootstrap_ci[2] * 100`%. This interval describes the sampling variability of the estimated hospitalization proportion under independent case-level resampling of the observed analytic cohort.

6.2 Refitting the Regression Model

Unlike the overall hospitalization proportion, the regression-standardized estimates depend on coefficients estimated from the observed data. If the composition of the analytic cohort changes, the fitted regression coefficients may also change.

As a result, each bootstrap replication must repeat the model-fitting procedure rather than reuse the coefficients estimated from the original cohort.

For each bootstrap replication, we draw a sample of n case records with replacement and refit the same logistic regression model. Repeating this procedure produces bootstrap distributions of the estimated regression coefficients.

These distributions reflect how the fitted coefficients vary across resamples of the observed analytic cohort, conditional on the resampling assumptions.

```
# Set seed for reproducibility
set.seed(42)

# Number of bootstrap replications
B_model <- 100

# Store medical-condition coefficient
bootstrap_beta <- rep(NA_real_, B_model)

# Exposure coefficient in original model
names(coef(logistic_model))
```

```
 [1] "(Intercept)"             "medcondYes"
 [3] "age_factor10 - 19 Years" "age_factor20 - 29 Years"
 [5] "age_factor30 - 39 Years" "age_factor40 - 49 Years"
 [7] "age_factor50 - 59 Years" "age_factor60 - 69 Years"
 [9] "age_factor70 - 79 Years" "age_factor80+ Years"
[11] "sex_factorMale"
```

```
exposure_term <- "medcondYes"

stopifnot(exposure_term %in% names(coef(logistic_model)))

bootstrap_models <- vector("list", B_model)
bootstrap_samples <- vector("list", B_model)

for (b in seq_len(B_model)) {

    # Draw case records with replacement
    indices <- sample(
        x = seq_len(n),
        size = n,
        replace = TRUE
    )

    # Construct bootstrap dataset
    boot_data <- adjusted_analysis[indices, , drop = FALSE]

    # Refit logistic regression model
```

```
    boot_model <- glm(
        hospitalized ~ medcond + age_factor + sex_factor,
        data = boot_data,
        family = binomial(link = "logit")
    )

    # Store exposure coefficient if model converged
    if (isTRUE(boot_model$converged) &&
    all(is.finite(coef(boot_model)))) {

        # Store the exposure coefficient
        bootstrap_beta[b] <- unname(coef(boot_model)[exposure_term])

        # Store the fitted model and its bootstrap sample
        bootstrap_models[[b]] <- boot_model
        bootstrap_samples[[b]] <- boot_data
    }
}

# Check for unsuccessful coefficient estimates
sum(!is.finite(bootstrap_beta))
```

```
[1] 0
```

```
# Summarize successful bootstrap coefficients
summary(bootstrap_beta[is.finite(bootstrap_beta)])
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.351   1.819   1.961   1.969   2.136   2.530
```

Each bootstrap replication produces a new logistic regression model fitted to a resampled version of the analytic cohort, with this analysis comprising 100 independent replications. Since observations may appear multiple times or be omitted from an individual resample, the estimated regression coefficients can differ from those obtained using the original data.

The coefficient associated with reported pre-existing medical conditions is retained from each fitted model. Its bootstrap distribution can be used to examine the variability of the estimated log odds ratio.

6.2.1 Examining the Bootstrap Coefficient Distribution

The histogram below displays the bootstrap estimates of the medical-condition coefficient on the log-odds scale. The vertical reference line indicates the coefficient estimated from the original analytic cohort.

```
# Check the actual coefficient names
names(coef(logistic_model))
```

```
 [1] "(Intercept)"              "medcondYes"
 [3] "age_factor10 - 19 Years" "age_factor20 - 29 Years"
 [5] "age_factor30 - 39 Years" "age_factor40 - 49 Years"
 [7] "age_factor50 - 59 Years" "age_factor60 - 69 Years"
 [9] "age_factor70 - 79 Years" "age_factor80+ Years"
[11] "sex_factorMale"
```

```
# Inspect bootstrap estimates
summary(bootstrap_beta)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.351   1.819   1.961   1.969   2.136   2.530
```

```
# Count valid estimates
sum(is.finite(bootstrap_beta))
```

```
[1] 100
```

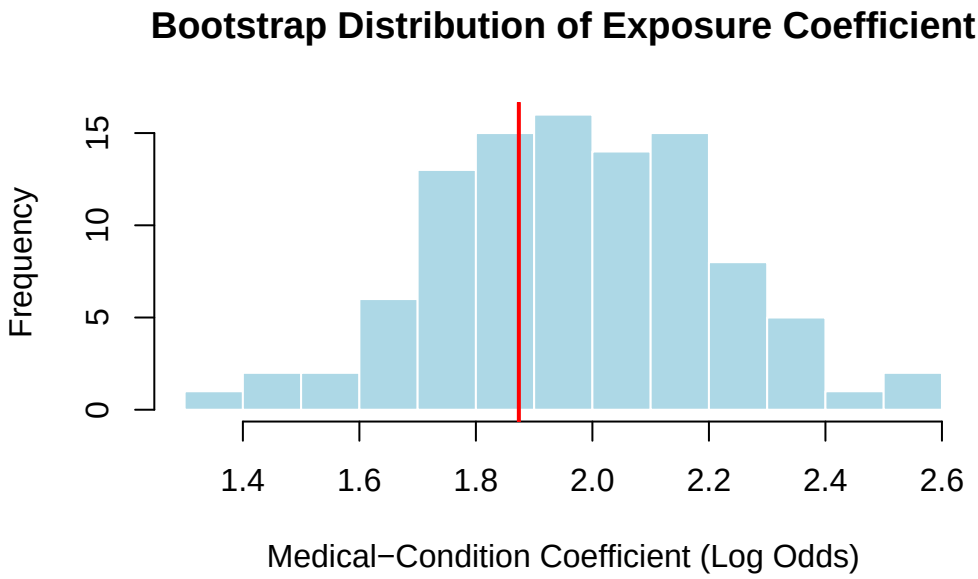
```
# Original medical-condition coefficient
original_beta <- unname(coef(logistic_model)[exposure_term])

# Retain successful bootstrap estimates
valid_beta <- bootstrap_beta[is.finite(bootstrap_beta)]

# Plot bootstrap coefficient distribution
hist(
    valid_beta,
    breaks = 15,
    main = "Bootstrap Distribution of Exposure Coefficient",
    xlab = "Medical-Condition Coefficient (Log Odds)",
    col = "lightblue",
```

```
    border = "white"
)

# Add original coefficient
abline(
    v = original_beta,
    col = "red",
    lwd = 2
)
```



The medical-condition coefficient estimated from the original logistic regression model was 1.874 on the log-odds scale. Across 100 bootstrap replications, the mean coefficient was 1.969, with a bootstrap standard error of 0.237.

The bootstrap estimates varied across resamples, demonstrating that the estimated conditional association between reported medical conditions and hospitalization is sensitive to the composition of the analytic cohort.

6.2.2 Recalculating Standardized Estimates

To extend bootstrap inference to the adjusted risk ratio (RR) and adjusted risk difference (RD), regression standardization must be applied to each of the 100 fitted bootstrap models.

For each bootstrap model, two exposure scenarios are constructed: one in which all cases are assigned reported pre-existing medical conditions and another in which all cases are assigned no reported pre-existing medical conditions. Age and sex remain unchanged.

Predicted hospitalization probabilities are averaged across the corresponding bootstrap sample to obtain standardized hospitalization proportions.

For bootstrap replication b, the standardized hospitalization proportion under exposure condition e is:

$$\hat{p}_e^{*(b)} = \frac{1}{n} \sum_{i=1}^{n} \hat{P}^{*(b)}(Y_i = 1 \mid E_i = e, A_i^{*(b)}, S_i^{*(b)}).$$

The adjusted risk ratio and risk difference are then calculated as:

$$\widehat{RR}^{*(b)} = \frac{\hat{p}_1^{*(b)}}{\hat{p}_0^{*(b)}}$$

$$\widehat{RD}^{*(b)} = \hat{p}_1^{*(b)} - \hat{p}_0^{*(b)}.$$

Repeating this procedure across the previously fitted bootstrap models produces empirical distributions of the standardized measures of association.

6.2.3 Calculating Standardized Estimates

```
# Store standardized estimates from existing bootstrap models
bootstrap_rr <- rep(NA_real_, B_model)
bootstrap_rd <- rep(NA_real_, B_model)

bootstrap_risk_exposed <- rep(NA_real_, B_model)
bootstrap_risk_unexposed <- rep(NA_real_, B_model)

# Use previously fitted models and their corresponding samples
for (b in seq_len(B_model)) {

    boot_model <- bootstrap_models[[b]]
    boot_data <- bootstrap_samples[[b]]

    # Skip unsuccessful model fits
    if (is.null(boot_model) || is.null(boot_data)) {
        next
    }

    # Construct exposure scenarios
```

```r
    exposed <- boot_data
    unexposed <- boot_data

    exposed$medcond <- factor(
        "Yes",
        levels = levels(adjusted_analysis$medcond)
    )

    unexposed$medcond <- factor(
        "No",
        levels = levels(adjusted_analysis$medcond)
    )

    # Predict hospitalization probabilities
    p_exposed <- predict(
        boot_model,
        newdata = exposed,
        type = "response"
    )

    p_unexposed <- predict(
        boot_model,
        newdata = unexposed,
        type = "response"
    )

    # Standardize over the bootstrap sample
    risk_exposed <- mean(p_exposed)
    risk_unexposed <- mean(p_unexposed)

    if (
        !is.finite(risk_exposed) ||
        !is.finite(risk_unexposed) ||
        risk_unexposed <= 0
    ) {
        next
    }

    # Store standardized proportions
    bootstrap_risk_exposed[b] <- risk_exposed
    bootstrap_risk_unexposed[b] <- risk_unexposed
```

```
    # Calculate adjusted measures
    bootstrap_rr[b] <- risk_exposed / risk_unexposed
    bootstrap_rd[b] <- risk_exposed - risk_unexposed
}

# Examine successful replications
sum(is.finite(bootstrap_rr) & is.finite(bootstrap_rd))
```

```
[1] 100
```

```
summary(bootstrap_rr)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2.415   3.559   4.109   4.189   4.733   7.005
```

```
summary(bootstrap_rd)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.1351  0.1820  0.1934  0.1931  0.2070  0.2508
```

Each successful bootstrap replication produces a standardized hospitalization proportion for both exposure scenarios, along with an adjusted RR and RD.

Due to the models being fitted to different resampled cohorts, the resulting estimates vary across replications. This variability reflects changes in the fitted regression coefficients and the empirical distribution of age and sex.

6.2.4 Examining the Bootstrap Distributions

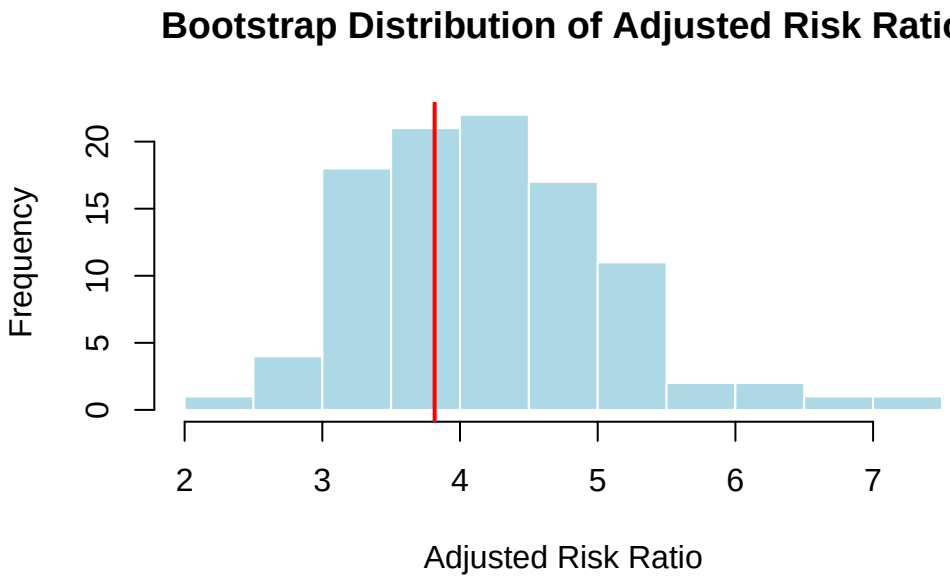
The distributions of the adjusted RR and RD can be examined to assess their center, spread, and shape.

Unlike the regression coefficient distribution, these distributions directly represent the measures of association used in the primary epidemiologic analysis.

```r
# Retain successful estimates
valid_rr <- bootstrap_rr[is.finite(bootstrap_rr)]
valid_rd <- bootstrap_rd[is.finite(bootstrap_rd)]

# Adjusted RR distribution
hist(
    valid_rr,
    breaks = 15,
    main = "Bootstrap Distribution of Adjusted Risk Ratio",
    xlab = "Adjusted Risk Ratio",
    col = "lightblue",
    border = "white"
)

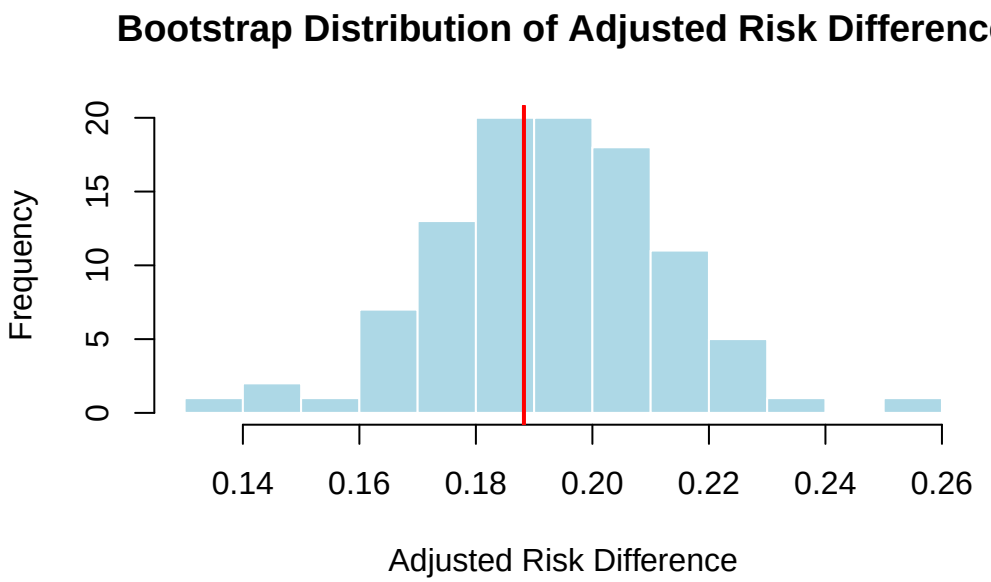
abline(
    v = adjusted_rr,
    col = "red",
    lwd = 2
)
```




```r
# Adjusted RD distribution
hist(
    valid_rd,
    breaks = 15,
    main = "Bootstrap Distribution of Adjusted Risk Difference",
```

```
    xlab = "Adjusted Risk Difference",
    col = "lightblue",
    border = "white"
)

abline(
    v = adjusted_rd,
    col = "red",
    lwd = 2
)
```



The red reference lines represent the adjusted estimates obtained from the original analytic cohort.

The bootstrap distributions illustrate how these estimates vary across resamples. The risk ratio and risk difference may exhibit different distributional shapes because they represent relative and absolute measures of association, respectively.

6.2.5 Bootstrap Confidence Intervals

The standard deviation of the bootstrap estimates provides an empirical estimate of the standard error.

The 95% percentile bootstrap confidence intervals are constructed using the 2.5th and 97.5th percentiles of each distribution.

```r
# Bootstrap standard errors
se_rr <- sd(valid_rr)
se_rd <- sd(valid_rd)

# Percentile confidence intervals
ci_rr <- quantile(
    valid_rr,
    probs = c(0.025, 0.975)
)

ci_rd <- quantile(
    valid_rd,
    probs = c(0.025, 0.975)
)

# Summary table
bootstrap_results <- data.frame(
    Measure = c(
        "Adjusted Risk Ratio",
        "Adjusted Risk Difference"
    ),
    Original_Estimate = c(
        adjusted_rr,
        adjusted_rd
    ),
    Bootstrap_Mean = c(
        mean(valid_rr),
        mean(valid_rd)
    ),
    Bootstrap_SE = c(
        se_rr,
        se_rd
    ),
    CI_Lower = c(
        ci_rr[1],
        ci_rd[1]
    ),
    CI_Upper = c(
        ci_rr[2],
        ci_rd[2]
    )
)
```

```r
knitr::kable(
    bootstrap_results,
    digits = 4,
    caption = "Bootstrap inference for regression-standardized measures"
)
```

Table 2: Bootstrap inference for regression-standardized measures

Measure	Original_Estimate	Bootstrap_Mean	Bootstrap_SE	CI_Lower	CI_Upper
Adjusted Risk Ratio	3.8157	4.1891	0.8726	2.7287	6.1987
Adjusted Risk Difference	0.1882	0.1931	0.0194	0.1495	0.2267

The regression-standardized adjusted risk ratio was 3.816 (95% percentile bootstrap CI: 2.73–6.2), indicating a positive adjusted association between reported pre-existing medical conditions and hospitalization.

The adjusted risk difference was 18.8 percentage points (95% percentile bootstrap CI: 14.9–22.7 percentage points).